Jieba dictionary training derivation

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October 12, 2026

1 Preliminaries and problem setup

`jieba` (`https://github.com/fxsjy/jieba`) is a well-known Chinese word segmentation library in Python. It comes with a default dictionary D of Chinese words w, each paired with a frequency score f_w. At inference time, it builds a directed acyclic graph (DAG) of dictionary words from every sentence. The edge $\mathsf{e}_{i\to j}$ in the DAG represents a dictionary word spanning from character i (inclusive) to character j (exclusive), with weight $s_{\mathsf{e}} \coloneqq \log f_{\mathsf{e}} - \log f_0$, where $f_0 = \sum_{w\in D} f_w$. Notice that we abuse the edge notation to also mean the corresponding dictionary word. Thus, for a sentence of length m, a path p the connects node 0 to node m defines a segmentation, and the sum of the edge weights $s_{\mathsf{p}} \coloneqq \sum_{\mathsf{e}\in\mathsf{p}} s_{\mathsf{e}}$ is the unigram log likelihood of a sequence of length $|\mathsf{p}|$.[1] Since the topological ordering of the nodes is known, we're able to decode the best segmentation p_* with the largest s_{p} efficiently by dynamic programming (DP). Some more details are:

- Out-of-vocabulary (OOV) words are split into characters $\mathsf{e}_{i\to i+1}$ and assigned edge weight $\log 1 - \log f_0$ when building the DAG. Hence, if there are many OOV words, the edge weights can no longer be interpreted as normalized log-probs.
- A hidden Markov model is optionally applied to OOV fragments to provide finer segmentation.

Our task is to augment the default dictionary D with a user dictionary U, and hope that the word segmentation quality is not impaired by the addition. We cannot assume the persistence of quality because U may contain frequency scores incompatible with D, e.g. one that originates from `https://github.com/iDvel/rime-ice`, and thus creating competing candidate paths in the DAG that negatively affects the decision. It's worth noting that word frequencies from U are sometimes adjusted in an ad hoc manner (see `https://github.com/fxsjy/jieba/issues`), which is inefficient and error-prone. Since $D \cup U$ is often sufficently large, we assume absence of OOV words thereafter to eliminate the cluttering of handling them.

[1] Strictly speaking, the edge weights sum is *not* the log likelihood of the segmentation p, as we'll explain in the next section.

2 A generative unigram model of sentences

Out of the DAG implementation, we prefer a different set of notations. In this section, we use the following notations: S for a segmentation, which is a sequence of dictionary words; x for an unsegmented sentence; $S_{\oplus}$ for the concatenation of the words in S, which itself is a sentence. We denote by D to refer the dictionary in general sense, since whether it's the default or the augmented dictionary usually does not matter. We also denote the parameters of a language model by $\boldsymbol{\phi}$. For example, $\boldsymbol{\phi}$ often includes the dictionary frequency scores f_w, whose normalized counterpart in terms of the dictionary is denoted by θ_w.

`jieba`'s segmentation log-prob $s_{\mathtt{p}}$, can be rewritten in our current parlance as $\log \prod_{w \in S} \theta_w$. We note that $\prod_{w \in S} \theta_w$ is *not* the likelihood $P(S; \theta)$ of observing S. It's instead the condition probability $P(S \mid |S| \text{ fixed}; \theta)$.

Theorem 1. *The sum of $P(S; \theta)$ over all possible S does not sum to 1.*

Proof.

$$\begin{aligned}
\sum_S P(S;\theta) &= \sum_{m=0}^{\infty} \sum_{S:|S|=m} P(S;\theta) \\
&= \sum_{m=0}^{\infty} \sum_{S:|S|=m} \prod_{w \in S} \theta_w \\
&= \sum_{m=0}^{\infty} \prod_{i=1}^{m} \sum_{w_i \in D} \theta_{w_i} \\
&= \sum_{m=0}^{\infty} \prod_{i=1}^{m} 1 \,,
\end{aligned}$$

and the r.h.s. diverges. Notice that as a byproduct we have shown that $\forall m$,

$$P(S \mid |S| = m; \theta) = \prod_{w \in S} \theta_w \,.$$

□

The reason is that there is no stopping rule in the generative story of `jieba`'s likelihood. For this purpose, there are two options: (1) define $\hat{D} \coloneqq D \cup \{\epsilon\}$ and stop generating new words after ϵ has been sampled, then $\hat{\theta}_w \propto \theta_w$; (2) for each token to be generated, first toss a biased coin whose head probability is τ to decide whether to stop, and then sample a token from D if we have decided to continue. It's straightforward to show that if $\tau = \hat{\theta}_\epsilon$, then $\hat{\theta}_w = \theta_w(1 - \tau)$, or in other words, the two stories are equivalent. Considering the fact that we ultimately need to return the unigram frequencies as output, parameterizing the stopping rule as an extra τ is more convenient, since it does not affect the normalization of θ_w's, neither does it affect how we will enforce prior on θ_w's.

2.1 Learning

Given a corpus of sentences C, we will train $\boldsymbol{\phi} = (\theta, \tau)$ by MLE/MAP of the log marginal likelihood. We will achieve this by EM.

The EM auxiliary function is

$$\begin{aligned} Q(\boldsymbol{\phi} \mid \boldsymbol{\phi}^{\text{old}}) &= \sum_{x \in C} \sum_{S} P(S \mid x; \boldsymbol{\phi}^{\text{old}}) \log P(S, x; \boldsymbol{\phi}) \\ &= \sum_{x \in C} \sum_{S} P(S \mid x; \boldsymbol{\phi}^{\text{old}}) (\log P(S; \boldsymbol{\phi}) + \log P(x \mid S)) \,. \end{aligned}$$

Since $P(x \mid S) = \mathbb{1}[x = S_{\oplus}]$ is irrelevant of $\boldsymbol{\phi}$, and the posterior $P(S \mid x)$ is only supported on region where $x = S_{\oplus}$, we can safely drop this term. Then,

$$Q(\boldsymbol{\phi} \mid \boldsymbol{\phi}^{\text{old}}) = \sum_{x \in C} \sum_{S} P(S \mid x; \boldsymbol{\phi}^{\text{old}}) \log P(S; \boldsymbol{\phi}) \,. \tag{1}$$

2.1.1 Learning θ

First, write:

$$\begin{aligned} \log P(S; \boldsymbol{\phi}) &= \log \left\{ \tau (1-\tau)^{|S|} \prod_{w \in S} \theta_w \right\} \\ &= \log \left\{ \tau \prod_{w \in S} (1-\tau) \theta_w \right\} \\ &= \log \left\{ \tau \prod_{w \in S} \hat{\theta}_w \right\} \\ &= \log \tau + \sum_{w \in S} \log \hat{\theta}_w \\ &= \log \tau + \sum_{w \in D} N_w(S) \log \hat{\theta}_w \,, \end{aligned}$$

where $N_w(S)$ denotes the number of occurrences of w in S. Plugging in to Equation (1) gives:

$$
\begin{aligned}
Q(\boldsymbol{\phi} \mid \boldsymbol{\phi}^{\text{old}}) &= \sum_{x \in C} \sum_{S} P(S \mid x; \boldsymbol{\phi}^{\text{old}}) \left(\log \tau + \sum_{w \in D} N_w(S) \log \hat{\theta}_w \right) \\
&= \sum_{x \in C} \sum_{S} P(S \mid x; \boldsymbol{\phi}^{\text{old}}) \sum_{w \in D} N_w(S) \log \hat{\theta}_w + \text{constant} \\
&= \sum_{w \in D} \left[\sum_{x \in C} \sum_{S} P(S \mid x; \boldsymbol{\phi}^{\text{old}}) N_w(S) \right] \log((1-\tau)\theta_w) + \text{constant} \\
&= \sum_{w \in D} \left[\sum_{x \in C} \sum_{S} P(S \mid x; \boldsymbol{\phi}^{\text{old}}) N_w(S) \right] \log \theta_w + \text{constant} \,.
\end{aligned}
$$

Hence,

$$
\theta_w \propto \sum_{x \in C} \mathbb{E}[N_w(S) \mid x; \boldsymbol{\phi}^{\text{old}}] \,,
$$

up to some additive offset from Dirichlet prior or pseudocounts.

2.1.2 Learning τ

Again, from Equation (1),

$$
\begin{aligned}
Q(\boldsymbol{\phi} \mid \boldsymbol{\phi}^{\text{old}}) &= \sum_{x \in C} \sum_{S} P(S \mid x; \boldsymbol{\phi}^{\text{old}}) \log \left(\tau(1-\tau)^{|S|} \prod_{w \in S} \theta_w \right) \\
&= \sum_{x \in C} \sum_{S} P(S \mid x; \boldsymbol{\phi}^{\text{old}}) (\log \tau + |S| \log(1-\tau)) + \text{constant} \\
&= |C| \log \tau + \left[\sum_{x \in C} \sum_{S} P(S \mid x; \boldsymbol{\phi}^{\text{old}}) |S| \right] \log(1-\tau) + \text{constant} \,.
\end{aligned}
$$

Taking derivative w.r.t. τ and making it zero yields:

$$
\tau = \frac{1}{1 + \tilde{m}} \,,
$$

where

$$
\tilde{m} = \frac{1}{|C|} \sum_{x \in C} \mathbb{E}[|S| \mid x; \boldsymbol{\phi}^{\text{old}}] \,.
$$

2.1.3 Finding the expectation in θ update

To evaluate $\mathbb{E}[N_w(S) \mid x]$, first notice that for S such that $S_{\oplus} = x$,

$$N_w(S) = \sum_{0 \le i < j \le |x|} \mathbb{1}[x_{i:j} = w \wedge (i,j) \in S] = \sum_{(i,j):x_{i:j}=w} \mathbb{1}[(i,j) \in S]$$

Hence,

$$\begin{aligned}
\mathbb{E}[N_w(S) \mid x] &= \sum_S P(S \mid x) \sum_{(i,j):x_{i:j}=w} \mathbb{1}[(i,j) \in S] \\
&= \sum_{(i,j):x_{i:j}=w} \sum_{S:(i,j)\in S} P(S) \Big/ P(x) \\
&= \sum_{(i,j):x_{i:j}=w} \sum_{S_p:(\cdot,i)\in S_p} \sum_{S_s:(j,\cdot)\in S_s} P(S_p \oplus (x_{i:j}) \oplus S_s) \Big/ P(x) \\
&= \sum_{(i,j):x_{i:j}=w} \sum_{S_p:(\cdot,i)\in S_p} \sum_{S_s:(j,\cdot)\in S_s} \tau \left(\prod_{w_1 \in S_p} \hat{\theta}_{w_1} \cdot \hat{\theta}_w \cdot \prod_{w_2 \in S_s} \hat{\theta}_{w_2} \right) \Big/ P(x) \\
&= \tau \sum_{(i,j):x_{i:j}=w} \hat{\theta}_w \sum_{S_p:(\cdot,i)\in S_p} \prod_{w_1 \in S_p} \hat{\theta}_{w_1} \sum_{S_s:(j,\cdot)\in S_s} \prod_{w_2 \in S_s} \hat{\theta}_{w_2} \Big/ P(x)\,,
\end{aligned}$$

where

$$P(x) = \tau \sum_{S:(0,\cdot)\in S \wedge (\cdot,|x|)\in S} \prod_{w \in S} \hat{\theta}_w\,.$$

Here, S_p denotes prefix segmentations that also satisifies $(0,\cdot) \in S_p$, and S_s denotes suffix segmentation that also satisfies $(\cdot,|x|) \in S_s$. Thus, this whole problem reduces to finding:

$$\tau^{-1} P(x_{s:t}) = \sum_{S:(s,\cdot)\in S \wedge (\cdot,t)\in S} \prod_{w \in S} \hat{\theta}_w\,.$$

and without loss of generality, let $s \coloneqq 0$ and $t \coloneqq n$. We may find $\tau^{-1}P(x_{0:n})$ by DP. Define $A_i \coloneqq \tau^{-1}P(x_{0:i})$. We need to find A_n. Then,

$$A_i \hat{\theta}_{x_{i:j}} = \sum_{S:(0,\cdot)\in S \wedge (\cdot,i)\in S} \prod_{w \in S \oplus (x_{i:j})} \hat{\theta}_w$$

And thus,

$$A_j = \sum_{i<j} A_i \hat{\theta}_{x_{i:j}}$$

2.1.4 Finding the expectation in τ update

To evaluate

$$\begin{aligned}\mathbb{E}[|S| \mid x] &= \sum_{S:(0,\cdot)\in S\wedge(\cdot,|x|)\in S} P(S \mid x)|S| \\ &= \sum_{S:(0,\cdot)\in S\wedge(\cdot,|x|)\in S} P(S)|S| \Big/ P(x)\,.\end{aligned}$$

As before, we define $A_i \coloneqq \tau^{-1}P(x_{0:i})$. We also define

$$B_i \coloneqq \tau^{-1} \sum_{S:(0,\cdot)\in S\wedge(\cdot,i)\in S} P(S)|S|\,.$$

We need to find $B_{|x|}$. Then,

$$\begin{aligned}A_i\hat{\theta}_{x_{i:j}} + B_i\hat{\theta}_{x_{i:j}} &= \tau^{-1} \sum_{S:(0,\cdot)\in S\wedge(\cdot,i)\in S} \prod_{w\in S\oplus(x_{i:j})} \hat{\theta}_w \\ &\quad + \tau^{-1} \sum_{S:(0,\cdot)\in S\wedge(\cdot,i)\in S} |S| \prod_{w\in S\oplus(x_{i:j})} \hat{\theta}_w \\ &= \tau^{-1} \sum_{S:(0,\cdot)\in S\wedge(\cdot,i)\in S} (|S|+1) \prod_{w\in S\oplus(x_{i:j})} \hat{\theta}_w \\ &= \tau^{-1} \sum_{S:(0,\cdot)\in S\wedge(\cdot,i)\in S} |S\oplus(x_{i:j})| \prod_{w\in S\oplus(x_{i:j})} \hat{\theta}_w\end{aligned}$$

Hence, $B_j = \sum_{i<j}(A_i + B_i)\hat{\theta}_{x_{i:j}}$.